

## CALAFAT, Romania

WMO: 154820

Lat: 43.9850N

Lon: 22.9461E

Elev: 204

StdP: 14.59

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 11.3       | 16.3 | 7.9                           | 8.3 | 12.5 | 13.4 | 10.9 | 19.4 | 34.7                  | 35.7 | 21.7 | 38.9 | 4.4                   | 260  | 0.450 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.9                   | 95.2            | 70.6 | 92.4 | 70.1 | 89.6 | 70.3 | 76.0                | 86.5 | 74.2 | 85.4 | 72.8 | 84.1 | 5.6                  | 120  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 72.9                            | 123.2 | 81.1 | 71.0 | 114.9 | 79.2 | 69.3 | 108.4 | 77.6 | 39.6          | 87.3 | 37.8 | 84.8 | 36.5 | 84.4 | 86.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |       |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|-------|-----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |       |     |
| (4) 22.4          | 18.1 | 15.6 | DB                         | 2.9 | 99.7               | 8.1 | 4.0                                                | -2.9 | 102.6      | -7.6 | 104.9      | -12.1 | 107.1      | -18.0 | 110.0 | (4) |
| (5)               |      |      | WB                         | 2.6 | 77.5               | 8.0 | 3.7                                                | -3.1 | 80.2       | -7.8 | 82.3       | -12.3 | 84.4       | -18.1 | 87.1  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 55.5   | 31.9  | 39.6  | 46.8  | 55.9  | 64.9  | 72.8  | 76.1  | 75.8  | 67.3  | 54.9  | 43.3  | 35.4  |  |  |
|                                                                        | DBStd       | 16.92  | 9.01  | 9.47  | 7.59  | 7.05  | 5.93  | 5.26  | 4.60  | 5.26  | 7.08  | 7.43  | 7.69  | 8.57  |  |  |
|                                                                        | HDD50       | 1763   | 563   | 306   | 148   | 27    | 0     | 0     | 0     | 0     | 0     | 37    | 226   | 456   |  |  |
|                                                                        | HDD65       | 4618   | 1027  | 711   | 565   | 280   | 77    | 6     | 1     | 4     | 61    | 318   | 652   | 916   |  |  |
|                                                                        | CDD50       | 3761   | 1     | 14    | 48    | 205   | 462   | 684   | 809   | 801   | 518   | 190   | 24    | 5     |  |  |
|                                                                        | CDD65       | 1143   | 0     | 0     | 0     | 8     | 74    | 241   | 345   | 340   | 128   | 7     | 0     | 0     |  |  |
|                                                                        | CDH74       | 11313  | 0     | 0     | 11    | 128   | 675   | 2293  | 3488  | 3476  | 1150  | 92    | 0     | 0     |  |  |
|                                                                        | CDH80       | 4886   | 0     | 0     | 0     | 26    | 149   | 930   | 1604  | 1719  | 444   | 14    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 5.5    | 4.7   | 6.4   | 6.7   | 7.0   | 5.7   | 5.3   | 5.3   | 4.8   | 4.9   | 4.8   | 4.7   | 5.5   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 22.1   | 1.5   | 1.4   | 1.5   | 1.9   | 2.2   | 2.1   | 2.1   | 1.7   | 1.9   | 2.0   | 1.7   | 2.0   |  |  |
|                                                                        | PrecMax     | 38.8   | 3.0   | 3.3   | 3.5   | 5.1   | 4.5   | 5.1   | 4.9   | 6.7   | 5.6   | 5.9   | 4.1   | 5.2   |  |  |
|                                                                        | PrecMin     | 10.9   | 0.3   | 0.0   | 0.2   | 0.0   | 0.9   | 0.4   | 0.2   | 0.0   | 0.1   | 0.2   | 0.0   | 0.0   |  |  |
|                                                                        | PrecStd     | 5.8    | 0.8   | 1.0   | 1.0   | 1.0   | 1.0   | 1.2   | 1.4   | 1.5   | 1.4   | 1.5   | 1.1   | 1.5   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 58.0  | 67.3  | 75.3  | 83.9  | 87.3  | 96.3  | 97.6  | 99.8  | 93.4  | 81.6  | 70.3  | 60.1  |  |  |
|                                                                        |             | MCWB   | 48.3  | 53.5  | 55.9  | 64.2  | 69.9  | 74.6  | 71.9  | 71.4  | 67.5  | 65.7  | 59.2  | 49.8  |  |  |
|                                                                        | 2%          | DB     | 51.4  | 61.6  | 68.5  | 77.5  | 84.0  | 91.8  | 94.5  | 95.7  | 89.0  | 76.1  | 63.1  | 54.8  |  |  |
|                                                                        |             | MCWB   | 44.7  | 50.9  | 53.6  | 63.0  | 67.2  | 71.7  | 70.9  | 70.0  | 67.8  | 62.2  | 54.9  | 47.2  |  |  |
|                                                                        | 5%          | DB     | 46.3  | 57.7  | 63.9  | 72.6  | 80.9  | 88.5  | 91.8  | 92.7  | 85.1  | 71.7  | 57.9  | 49.7  |  |  |
|                                                                        |             | MCWB   | 41.8  | 49.7  | 51.3  | 59.3  | 65.6  | 71.2  | 70.7  | 69.4  | 66.6  | 59.9  | 52.6  | 43.9  |  |  |
|                                                                        | 10%         | DB     | 42.7  | 53.4  | 59.5  | 68.9  | 77.8  | 85.5  | 88.6  | 89.5  | 81.1  | 67.7  | 53.6  | 46.4  |  |  |
|                                                                        |             | MCWB   | 39.8  | 47.2  | 49.0  | 57.4  | 64.5  | 70.2  | 70.9  | 69.5  | 66.0  | 59.2  | 50.9  | 41.9  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 48.7  | 56.3  | 57.9  | 66.3  | 74.8  | 77.2  | 78.5  | 78.6  | 75.7  | 70.5  | 59.2  | 51.2  |  |  |
|                                                                        |             | MCDB   | 56.6  | 62.4  | 69.1  | 78.3  | 85.0  | 89.2  | 87.1  | 90.1  | 86.2  | 73.8  | 67.0  | 56.8  |  |  |
|                                                                        | 2%          | WB     | 45.4  | 52.8  | 54.8  | 64.0  | 70.3  | 74.6  | 76.3  | 74.7  | 72.0  | 63.6  | 56.0  | 47.9  |  |  |
|                                                                        |             | MCDB   | 50.3  | 59.6  | 66.9  | 75.7  | 79.4  | 86.3  | 85.1  | 85.9  | 82.4  | 71.9  | 61.2  | 54.4  |  |  |
|                                                                        | 5%          | WB     | 42.6  | 50.1  | 52.4  | 61.3  | 67.9  | 73.3  | 74.2  | 73.0  | 69.1  | 61.1  | 53.7  | 44.9  |  |  |
|                                                                        |             | MCDB   | 45.2  | 56.6  | 62.5  | 71.0  | 77.5  | 84.3  | 85.1  | 85.4  | 79.1  | 70.3  | 57.1  | 49.0  |  |  |
|                                                                        | 10%         | WB     | 40.3  | 47.3  | 50.2  | 58.9  | 65.5  | 71.9  | 72.5  | 71.3  | 66.8  | 59.1  | 51.1  | 42.5  |  |  |
|                                                                        |             | MCDB   | 42.8  | 52.6  | 58.3  | 66.8  | 74.7  | 82.3  | 84.3  | 83.6  | 76.8  | 66.9  | 53.5  | 45.8  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.6  | 14.2  | 17.9  | 19.3  | 19.6  | 21.3  | 21.9  | 23.9  | 21.1  | 18.0  | 10.9  | 11.3  |  |  |
|                                                                        |             | MCDBR  | 18.5  | 21.8  | 27.0  | 27.4  | 25.3  | 26.7  | 27.9  | 29.8  | 27.8  | 26.4  | 19.5  | 17.5  |  |  |
|                                                                        |             | MCWBR  | 12.9  | 13.9  | 14.7  | 13.9  | 11.2  | 9.7   | 8.6   | 9.1   | 9.9   | 13.4  | 12.3  | 11.6  |  |  |
|                                                                        |             | MCDBR  | 16.4  | 20.2  | 24.5  | 24.8  | 22.9  | 23.9  | 22.3  | 24.1  | 24.0  | 24.2  | 15.7  | 16.1  |  |  |
|                                                                        | 5% WB       | MCWBR  | 12.0  | 13.9  | 14.1  | 13.5  | 11.7  | 10.1  | 9.9   | 10.3  | 10.8  | 13.6  | 10.8  | 11.4  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.346 | 0.374 | 0.413 | 0.474 | 0.454 | 0.448 | 0.459 | 0.477 | 0.446 | 0.403 | 0.372 | 0.346 |  |  |
|                                                                        | taud        |        | 2.288 | 2.204 | 2.168 | 2.019 | 2.143 | 2.200 | 2.170 | 2.117 | 2.168 | 2.278 | 2.315 | 2.312 |  |  |
|                                                                        | Ebn at Noon |        | 239   | 252   | 258   | 251   | 261   | 263   | 258   | 248   | 245   | 240   | 228   | 227   |  |  |
|                                                                        | Edh at Noon |        | 27    | 35    | 42    | 52    | 47    | 45    | 46    | 47    | 41    | 32    | 27    | 24    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 448   | 706   | 1075  | 1453  | 1773  | 1996  | 2014  | 1793  | 1284  | 837   | 484   | 376   |  |  |
|                                                                        | RadStd      |        | 66    | 102   | 136   | 165   | 155   | 164   | 134   | 155   | 110   | 117   | 68    | 54    |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (55 neighbors) |  | +1.06 | N/A     | N/A    | N/A     | +0.72 | N/A   | N/A         | -268  | +167  | N/A   |

Nomenclature: See separate page